Predicting Criminal Sentencing Outcomes from Neighborhood Income and Type of Defense Counsel in Harris County, Texas

COSC 3337 Data Science I, Term Project Proposal | Instructor: Pam Qian
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1. Introduction and Motivation

Sentencing should depend on what someone did, not on where they live or who represented them. Research on sentencing keeps finding otherwise, reporting gaps by race and income that remain after accounting for the offense itself [1, 2, 4]. Anderson and Heaton [3] went further and showed that the type of defense counsel a defendant gets changes the outcome of the case.

Most county court records do not let anyone test that, because they say nothing about counsel or about where the defendant lives. Harris County is an exception. Its public disposition records list the defendant's address and whether their attorney was hired, appointed, or a public defender, which lets us ask: **once we account for the offense charged, do neighborhood income and type of counsel help predict how severe the sentence is?** Counsel assignment is something the county controls, so if it carries predictive weight, that is a finding someone can act on.

2. Proposed Method

We will build two models and compare them on the same data.

Traditional method (covered before Midterm 1): multinomial logistic regression, with a decision tree as a second baseline. The target is a category and most of our predictors are categories, which fits logistic regression well. It also gives us coefficients we can read directly, so we can say which factors push sentences in which direction. For a project about fairness, being able to explain the model matters as much as its accuracy.

Neural network: a small multilayer perceptron, around two hidden layers of 32 to 64 units, with dropout and early stopping. An MLP can pick up combinations the linear model cannot, for example appointed counsel in a low-income tract being worse than either factor on its own. Comparing the two tells us whether those combinations actually matter here.

Target: sentence severity, grouped into probation or no jail time, jail measured in days, and prison measured in years. **Features:** offense level and degree, offense category, bond amount, court, defendant age, sex and race, type of defense counsel, and census-tract income from the ACS.

These choices reflect what we have covered in the course so far. We expect to revise them as the semester introduces new methods, and we may swap in or add a model if a better fit for this data comes up later.

3. Evaluation

We will split the cases 70 percent training, 15 percent validation and 15 percent test, stratified so each split has the same mix of outcome classes. The test set stays untouched until the end. Preprocessing (imputation, encoding, scaling) is fit on training data only, so nothing leaks into validation or test, and tuning uses 5-fold cross-validation inside the training set. The classes are uneven, so plain accuracy would look good while the model ignored rare outcomes. We report **macro-averaged F1, balanced accuracy, and confusion matrices** instead, with precision and recall per class. Both models get the same folds and test set. We will also report results separately by counsel type and tract income group, since an average can hide uneven performance across the groups we are studying, and refit both models without the socioeconomic features to see how much they add over offense details alone.

4. Datasets

(a) Harris County District Clerk, criminal disposition records. Source: hcdistrictclerk.com, Public Datasets, Criminal section [5]. The clerk publishes tab-delimited bulk files daily, monthly, and as five-year archives going back to 1990. We will use the 2020 to 2024 archive. Each record has 36 fields. To check the structure we looked at the file published 1 November 2026, covering October 2026 dispositions. It has 11,967 rows but only 6,692 unique cases, because a case repeats once per charge, so deduplication is a required first step. On those 6,692 cases, disposition and offense level are filled on 6,684, counsel type on 6,202, ZIP on 5,842 and sentence on 974. Counsel is 1,833 appointed, 1,723 hired, 1,418 MAC appointed and 1,225 public defender, with 490 not recorded. One month gives too few sentenced cases to train on, hence the five-year archive. Scaling that month across 60 months suggests roughly 400,000 dispositions and on the order of 50,000 to 60,000 sentenced cases as our modeling set, with about 13 features before encoding, 9 from the case record and 4 from the tract join. That is an estimate from one month, to be confirmed after we deduplicate the archive, and 2020 to 2021 court volumes were disrupted by pandemic backlogs, so the true count may be lower. **Prediction target: sentence severity.**

(b) American Community Survey 2024 5-Year Estimates, census tract level. Source: data.census.gov [6]. We have already pulled median family income (table B19113) for all **1,115 Harris County tracts** and joined it to TIGER/Line 2024 tract boundaries [7]. We will add household income (B19013), the Gini index (B19083) and poverty rate (S1701) on the same join. Defendant addresses will be geocoded to tracts with the Census Geocoder, falling back to ZIP-level data if too few addresses match.

5. Team Contributions

Olajide. Data and geospatial pipeline: downloading and parsing the clerk archives, deduplication, geocoding addresses, the ACS and TIGER join, and the tract feature layer. Writes the Datasets and Proposed Method sections and keeps the shared repository.

Aidan. Traditional models: feature encoding, logistic regression and the decision tree, the cross-validation setup, and interpreting coefficients. Writes Related Work and Experiments.

Diego. Neural network: MLP architecture and training, regularization and early stopping, plus the shared evaluation code including the sliced metrics and confusion matrices. Writes Results and Discussion and Conclusions.

All three of us review the full report, and we split exploratory analysis and figures evenly across the three tracks.

References

[1] Spohn, C. (2000). Thirty years of sentencing reform: The quest for a racially neutral sentencing process. *Criminal Justice 2000*, 3, 427-501.

[2] Mitchell, O. (2005). A meta-analysis of race and sentencing research: Explaining the inconsistencies. *Journal of Quantitative Criminology*, 21(4), 439-466.

[3] Anderson, J. M., and Heaton, P. (2012). How much difference does the lawyer make? The effect of defense counsel on murder case outcomes. *Yale Law Journal*, 122(1), 154-217.

[4] Ulmer, J. T. (2012). Recent developments and new directions in sentencing research. *Justice Quarterly*, 29(1), 1-40.

[5] Harris County District Clerk. Public Datasets. https://www.hcdistrictclerk.com/Common/e-services/PublicDatasets.aspx

[6] U.S. Census Bureau. American Community Survey 2024 5-Year Estimates. https://data.census.gov

[7] U.S. Census Bureau. TIGER/Line Shapefiles 2024, Census Tracts. census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html